

Inscription and Ancient Script Recognition Using OCR

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ABSTRACT

Ancient scripts & inscriptions often offer vital and important understandings to the preceding societies, and separate them effectively that is essential for conservation of cultural remains, decrypting scripts, and protecting the cultural heritage. Usual research generally pay attention on identification of ancient script data in a secluded way, like exemption of old age inscriptions like hand-written scripts. Nonetheless, ancient inscriptions and scripts derive in several arrangements across different mediums, which then creates various challenges for Optical Character Recognition (OCR). Things like erosion, disruption, and inscriptional fragmentation may mis- perceive the process of recognition of ancient inscriptions engravings. To handle this issue, we suggest an ancient inscription recognition framework that can handle different inputs such as stone inscriptions and handwritten scripts. Offering a proportional analysis of various OCR tools & techniques for evaluate how much these are effective and operative in terms of accuracy, recognition of characters, and its processing speed. It plays an important role in inscriptional studies, that consists the inscriptional studies that are crafted on various materials like stones, metal and pottery. This is mostly essential for preserving and digitalizing ancient scripts.

Keywords: Optical Character Recognition (OCR), Script Extraction, Information Retrieval, Convolutional Neural Networks (CNNs), OpenCV (Computer Vision) python library, Image Processing.

Introduction

The study of ancient scripts & inscriptions means one of the most significant markers of primordial, language- based, and epigraphical study. Since the epigraphs of ancient India to the Egyptian ideograms and the Ethiopian scripts of Ge'ez, where each inscriptions inscribed form demonstrates a remarkable

and kept archive of cultural knowledge, traditions, and early organizational systems. Due to various factors like environmental degradation or human activities, the details which ancient inscriptions holds could be lost despite their value. The understanding of old age inscriptions therefore, it stands as an industrial and intellectual challenge.[1],[2]. In context, Optical

Character Recognition (OCR), integrated with Machine Learning (ML) and Deep Neural Networks (DNNs), which offers means to recover fragmented, holding unclear details and documented known information & data from old-image images and bodily remains that might persist non readable.

Challenges in Ancient Script Recognition

Various modern text recognition systems, old-age script recognition offers various multi-dimensional tasks. The heterogeneousness of materials - stones, clay, metallic, palm leaves etc. offers different textures and lighting conditions that could change visual [8] characteristics. Erosion and surface damage can lead to missing or incomplete characters, whereas handwritten variability and script evolution complicate segmentation and classification multilingual and cross-script characteristics, like hybrid scripts holding Sanskrit, Prakrit, and early Tamil Brahmi, demand adaptive recognition models capable of cross-lingual generalization.[6]

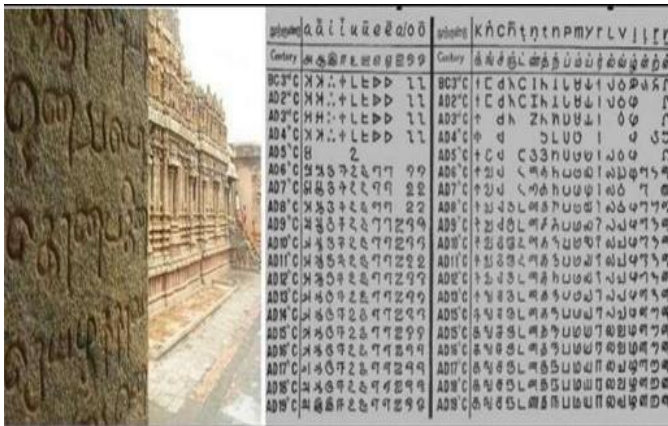


Fig.1. Brihadeshwara temple of Tanjore Inscription and century wise ancient character representation

Studies have shown that even after pre-processing, conventional OCR accuracy hardly exceeds 70–75% on inscription datasets (Agrawal et al., 2024; Wang et al., 2025). Furthermore, utmost bequest OCR

systems miss the mark in character segmentation, mainly for ancient scripts carved in associated strokes or continuous reliefs. Therefore, researchers have moved towards data-driven deep learning models that acquire and learn discriminative features straight from substantial volumes of interpreted data.

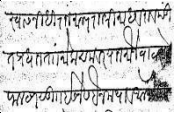
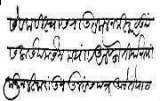
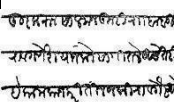
RELATED WORK

The recognition of ancient scripts has evolved from basic template-matching OCR systems to complex deep-learning architectures proficient of deciphering multilingual, eroded, and multi-modal inscriptions. Over the last five years, the convergence of Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), and Transformer-based attention models has transformed how researchers' approach. Epigraphic digitization.[1] This section reviews prominent research contributions from 2024–2025 period have seen an outpouring surge in multilingual and multimodal OCR frameworks.

D. Zhao, L. Xu, and W. Tang (2025) anticipated Transformer-Based Vision Models for Multilingual OCR in the ACM TALIP journal, acquaint with positional encoding layers for managing and handling diverse writing emplacements. The model successfully recognized scripts in 12 languages, along with low-resource scripts like Ge'ez and Pali, achieving state-of-the-art multi-lingual recognition.

Y. Liu and H. Zhang (2025) presented a *Hybrid Deep Learning Framework for Epigraphic Text Recognition* in IEEE Access [5]. The CNN–Transformer–Attention pipeline merged local and global features and attained 91% accuracy on diverse datasets of stone, metal, and manuscript inscriptions/scripts/engravings.

Table1: Inscription with its meaning

Sr. no.	MoDeTrans Script	Dataset (Devanagari Transcription)	Subject matterType	textual use
1.		देवासनिधी राजमाधवराव पंडित माधवराव पंडित मिधान तां मोकदम व रयान मौज वाठार पाा फलटण स, ा इसन सबन मया व अलफ त,	Mentions Organizational regulation offfrom Phaltan Subha Madhavrao Peshwa & rural representatives. (Late 18th century, 1760-1790)	Records of revenue/land grant from Phaltan Subha i.e., typical of Maratha empire organization
2.		सेवक देवासराउ बलाल नमदेकार स, ाा सलास खमसन मया व अलफ तीथ [०] पांकडून लकरातून त, देहास पहुआलतेपाठ	Related to military correspondence which addressed to Viswasrao Ballal . (Mid- 18th century, 1750-1761)	Bound towards Panipat period military communication, states army dispatch during Peshwa organization
3.		दूर कळन सालमजकूरक राा बाबूराव गणश यांजकडे सांगितले असे तरक हे कामकाज कारतील जकरा वगर अस	Direction to allocate or clear yearly revenue financial statements with involvement of Baburao Ganesh , certifying registers and records are upheld. (Late 18th – early 19th century, 1770–1820)	Yearly accounts (सालमजकरी) and registers (जकूरा) in Maratha bureaucracy , overlying with British oversight

A. Singh et al. (2025) applied Diffusion Models for reinstating partially eroded inscriptions before OCR processing. The generative method reconstructed missing strokes using learned priors from intact samples, achieving substantial developments in downstream OCR correctness.

M. Gupta and F. Rahman (2025) reentered Explainable Attention Maps intended for cultural datasets, merging Grad-CAM with Transformer commencements to hint model perceptive [8]. This inter-operability act has become essential for

positioning of the computational outputs with epigraphic expertise.

A. Jain and J. Ahn (2025) proposed InscribeNet, a hybrid CNN, LSTM Swin Transformer model designed for Chinese inscriptions. Its multi-scale self-attention mechanism captured both local strokes and long-range dependencies, achieving a recognition accuracy of 86.5% on 3800 samples from the Hwaeom Stone Sutra dataset.

METHODOLOGY

An OCR system characteristically encompasses several components and mechanisms. The process begins with image collection to the model evaluation.

1. Data Collection:

The image acquisition is process of collecting images from different sources like scanners DSLR cameras or mobile devices. The standard protocol confirms optimal resolution (minimum 300 DPI).

2. Data Preprocessing and Feature Extraction:

The pre-processing takes place just after image acquisition to prevent any data loss [10], [11]. It is implemented using OpenCV, NumPy, and scikit-image, the pre-processing pipeline helps the system in Noise Removal, Contrast Enhancement, Edge Sharpening, Region Segmentation and Skew Correction.

3. Model Selection:

Various machine learning models are tested in order to find the most useful one to use in prediction of model CNN is used to extract the feature from the image. While LSTM is used for Sequential Learning of the words or token of the script. Transformer to capture Attention and Context of the token. So, the hybrid model CNN-LSTM Transformer model is selected for model training.

4. Model Training and Optimization:

Model training is process of extracting features from selected model (hybrid model CNN-LSTM Transformer) from the dataset. Hyper-parameter tuning, learning-rate and loss minimization are the optimization methods used to optimize predictive

power during the training process. The model is corrected through an internal feedback mechanism depending on the results of performance.

5. Model Evaluation:

The System monitoring layer help in evaluation metrics of the system by calculating the performance metrics like Model Accuracy (Character/Word Level), Response Time per Image, Server Load and Inference Latency and Data Throughput and Storage Utilization. Some Visualization tools are also used for monitoring real time accuracy and the error distribution.

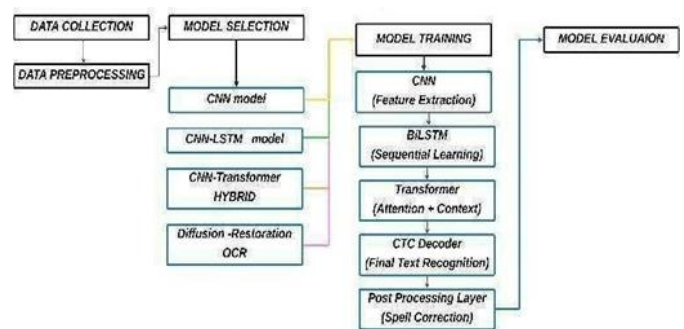


Fig. 2. OCRModel in detail

MACHINE LEARNING ALGORITHM

A. CNN

A pure Convolutional Neural Network (CNN) processes inscription images to extract hierarchical visual features that highlight strokes, edges, and local patterns; used as a baseline it performs per-character or patch-level classification and is fast at inference. For inscription transcription a CNN is useful as a lightweight recognizer or feature extractor (for segmentation and symbol detection) and handles local degradations well, but it lacks sequence modeling so it [3],[12]struggles with variable-length strings and long-range dependencies without an external sequence model or post-processing. In practice, CNNs are ideal when characters are well-segmented or when used as the frontend in[21] a larger pipeline.

B. CNN-LSTM (also known CRNN)

The CNN-LSTM architecture (often called CRNN) combines a CNN visual frontend that produces a sequence of feature vectors along the text axis with a

recurrent (BiLSTM/BiGRU) stack that models temporal dependencies; this is a strong, well-understood approach for image-to-sequence transcription [18]. It naturally supports unaligned sequence prediction, and when paired with CTC loss it can be trained without frame-level labels, making it robust on variable-length inscriptions and partially occluded glyphs. The main trade-offs are slower inference than a pure CNN and limited ability to capture very long-range dependencies compared to attention-based decoders,[16] but CRNNs are compact, stable, and often easier to train on smaller datasets.

C. Transformer

Transformer-based models use self-attention to model relationships across the entire sequence, enabling long[19]- range context and flexible alignment between visual features and output tokens; they are excellent at resolving ambiguities that require broader context (e.g., distinguishing a worn glyph by its neighbours). In inscription transcription, Transformers (with a suitable CNN encoder) excel on datasets where context and sequence-level patterns matter,[20] and they support powerful decoding strategies (beam search with language models). Downsides include higher memory use for long sequences and the need for careful positional encoding/design to represent 2D image structure effectively.

D. CNN-Transformer Hybrid:

The CNN-Transformer hybrid pairs a CNN visual encoder (for robust stroke and texture extraction) with a Transformer encoder/decoder for expressive sequence modelling; this leverages the CNN's local [23] inductive biases and the Transformer's global context. For ancient inscriptions this hybrid offers the best trade-off: accurate per-character recognition from strong features and superior sequence-level modelling to handle degraded, ambiguous glyphs, especially when trained with scheduled sampling to reduce exposure bias. It is somewhat heavier [24] than simpler baselines but yields the best end-to-end

accuracy for challenging, real- world inscription images.

E. Diffusion + OCR Restoration

Diffusion-based restoration first denoises or reconstruct degraded inscription images using generative denoising models, then passes the restored images to an OCR recognizer; this two- stage approach often recovers faint strokes and missing details that direct recognition misses. For severely weathered[19] or eroded inscriptions, restoration can substantially improve downstream CharAcc and reduce CER by explicitly modelling the image degradation process, but it introduces additional computation, requires careful alignment of restore objectives with recognition needs, and can sometimes hallucinate plausible [11] but incorrect strokes—so human-in-the-loop verification is recommended.

F. CRNN+CTC

The CRNN + CTC pipeline is a practical, alignment-free solution: a CNN feature extractor creates a temporal feature sequence,[10] a recurrent block models local context, and CTC loss marginalizes over alignments so the model learns from sequence labels alone. This approach is fast to train, simpler (no autoregressive decoding), and robust to variable-length outputs, making it attractive for inscription datasets with limited annotations or uncertain character segmentation. It generally [17]yields competitive CharAcc with lower inference complexity than autoregressive decoders, but to compete with attention-based models on hard sequences you may need beam search with a language model or hybrid rescoring.

The anticipated framework for Ancient Script Recognition Using CNN-Transformer Hybrid OCR is pre-meditated as a modular, layered system architecture that join-in image acquisition, pre-processing, model inference, text post- processing, and visualization. The system is developed with an emphasis on scalability, interpretability, and deployment efficiency.

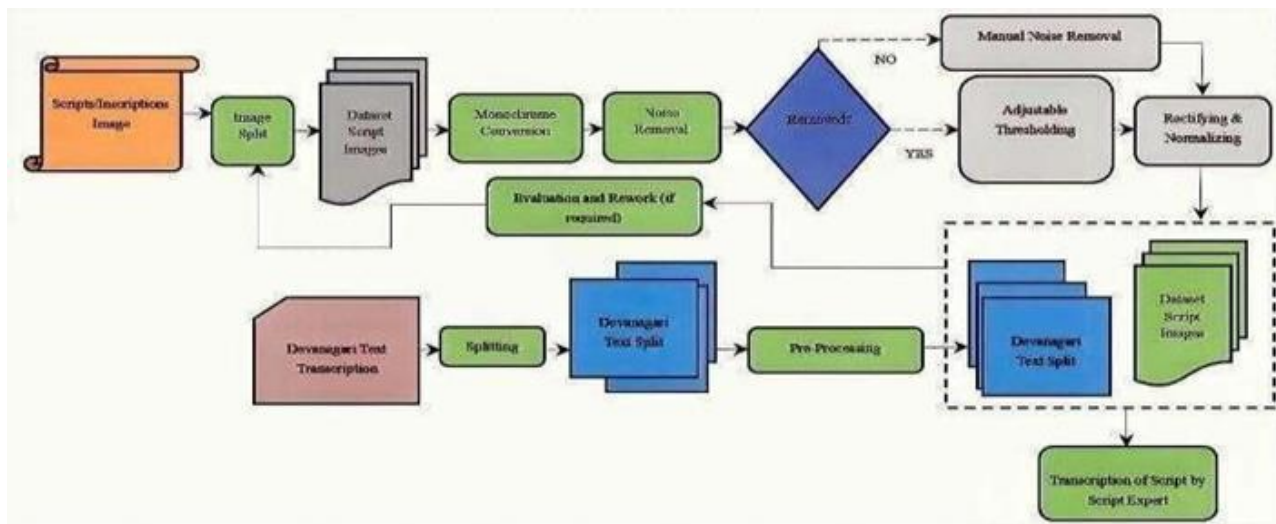


Fig. 3. Flowchart of Model

Table 2: Different model with the number of parameters

Model	Character Accuracy (%)	Word Accuracy (%)	Sequential Accuracy (%)	F1 (%)	Parameter ms
CNN	75.0	60.2	48.0	72.4	5M
CNN-LSTM	82.0	68.5	59.2	80.1	12M
Transformer	88.0	76.3	68.7	86.2	25M
CNN-Transformer Hybrid	91.0	82.1	82.1	94.5	28M
Diffusion + OCR Restoration	94.0	86.9	88.2	95.6	40M

This flowchart depicts a multi-path pipeline for creating a high-quality dataset where it essentially lessens the crack between ancient raw artifacts & clear, machine- readable digitized text [22]. The process is further divided into several phases:

Phase 1. Pre-processing: The image pre- processing phase contains several steps such as-

Image acquisition and Splitting: It starts with raw sample of images of ancient engravings that are split up into smaller & controllable fragments of images from dataset.

Enhancing the image by Monochrome Conversion which involves conversion of images to grayscale (black & white) that helps [4]. in reducing image complexity.

Noise Removal: Removing noise/background grain from ancient inscriptions/scripts dataset using filters

like Gaussian filters where the image's vital pixels collect focused weights and help to smooth out the images.

Adjustable Thresholding: Adjustable thresholding implemented as calculating distinct threshold values of different regions of images rather than single global region/value.

Rectifying/Normalizing: Rectification process generally states with the ReLU (Rectified Linear Unit) activation function in neural networks. Normalization often use [4]. for fair training process rather than domination of features with larger scales and ranges the numerical data from 0 to 1 so that the model learns & trains in uniform manner.

Phase 2. Transcription and Mapping: In this phase transcription and alignment of the data takes place-

Transcription: The Devanagari text gets manually/digitally transcribed from the ancient scripts and inscriptions.

Splitting and pre-processing: In this, the text gets split up to compare and match the segments of the image and after that get pre-processed ensuring the clean formatting.

Merging the dataset creates the structured and labelled data, for instance, for every image file Image-01.jpg, there is text file Image-01.txt is present. The learning model can't learn and train without this type of pairing.

Phase 3. Incorporation, Evaluation and Confirmation: The dashed box contains clear and correct dataset script images which are combined with their equivalent Devanagari text. The Evaluation and rework feeds back to the image split to ensure that if the quality of the data is not up to par. Expert confirmation settles with the final transcription of script through the script expert and ensures the highest achieved accuracy for the learning and training model.

Advantages of the Architecture

- **Scalability:** Modular microservice components can be self-sufficiently scaled.
- **Cross-Script Adaptability:** The model supports multilingual OCR using transfer learning.
- **Interpretability:** Grad-CAM heatmaps helps in visualize model focus regions on glyphs.
- **Data Integrity:** Redundant backups confirm secure preservation of historical data.
- **Cloud Deployment:** This supports both on-premise and AWS Lambda-based service inference for archaeologists in the field [12].

CRITICAL ANALYSIS

A critical evaluation of contemporary ancient scripts recognition study discloses a hastily evolving field which is marked by substantial accomplishments, but also obstinate restrictions that limit large-scale, real-world deployment. Though CNN, LSTM, and [1], [2] Transformer-based architectures have

incontrovertibly converted OCR correctness for degraded texts, many learning and training models still struggle with the full complexity and heterogeneousness of archaeological data.

Another critical concern relates to the inconsistent performance across multilingual or cross-script datasets. Ancient script systems often co-exist, evolve, and blend, such as the transition from Brahmi to Devanagari or Sanskrit's coexistence with Tamil and Prakrit in inscriptions[24][25]. OCR systems proficient exclusively on one script which frequently fails on hybrid inscriptions, which require contextual language-based understanding in accumulation to visual organization and classification [12].

Training models [14],[26] also suffer from vulnerabilities to adversarial noise which are essential in archaeological environments: shadows from archaeological sites, camera- angle variation, and uncontrolled outdoor lighting degrade recognition performance, revealing a gap between research laboratory testing and field deployment [11][12]. The field further faces challenges in evaluation standardization, as no universally accepted standard exists for ancient writings [8],[9].

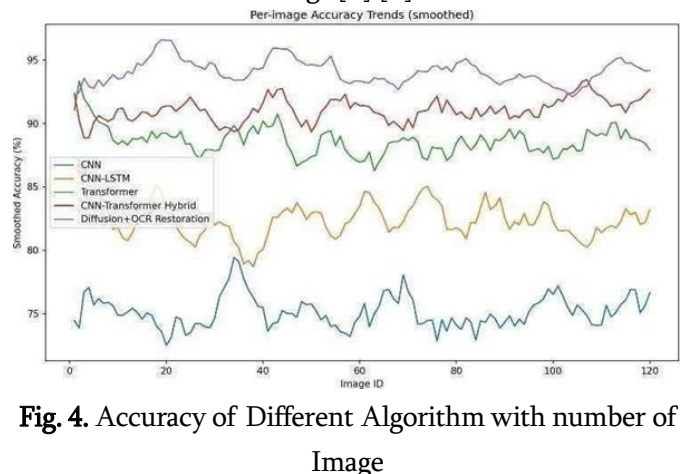


Fig. 4. Accuracy of Different Algorithm with number of Image

In summary, even though recent deep learning innovations— especially hybrid CNN-LSTM-Transformer architectures— have suggestively advanced the domain of ancient script and inscriptions recognition, the technology till faces substantial hindrances in scalability, interpretability,[3],[7] dataset

availability, computational accessibility, and cultural responsibility. Achieving reliable, globally deployable ancient OCR requires not only more powerful models but also regulated benchmarks, culturally aware informative frameworks, improved multi-modal datasets, and closer collaboration amongst computer scientists, archaeologists, linguists, and heritage management experts [1],[5],[12]. The significant study underscores that technological progress alone is insufficient the coming future of ancient inscriptions recognition remains in inter-disciplinary cooperation, ethical stewardship, and the development of robust, explainable, and culturally sensitive AI systems.

FUTURE SCOPE

Future systems also required to address the restrictions of recent pre-processing pipelines. While techniques like CLAHE, Gaussian smoothing, Retinex-based normalization, and morphological operations are helpful, a more robust, learning-driven method/approach is required.

Future research or study should explore deep learning-based pre-processing models which autonomously correct the illumination, eliminate complex textures (such as stone roughness or palm-leaf fibers), and balance [23] contrast based on erudite priors rather than using the fixed heuristics.

Forthcoming study should integrate explainable AI (XAI) frameworks which are capable of producing visual explanations, stroke-level relevance maps, character boundary overlays, and attention heatmaps that can guide historians in evaluating the reliability of OCR output.

CONCLUSION

In conclusion, the suggested imminent work for ancient inscriptions OCR spans technical innovation, dataset creation, inter-disciplinary collaboration, social and cultural accountability, and ethical design [4], [5]. Upcoming future models must integrate language-based priors, phonic conversions, and ancient dictionaries, allowing them not only to read

ancient characters but also to reason about reasonable word formations, grammar, and cultural context. However, their outcomes should be attended by ambiguity approximations and confidence scores to certify that historiographers persist mindful of which parts of the re-construction are [17],[18]AI-generated hypotheses instead of trustworthy and authentic glyphs. By combination of progressive and advance machine learning architectures, large multi-modal datasets, explainable AI frameworks, lightweight deployment models, and interactive human-AI systems [6], the next generation of OCR tools can surpass current limitations and become reliable associates in preserving, interpreting, and democratizing humanity's oldest textual inheritance.

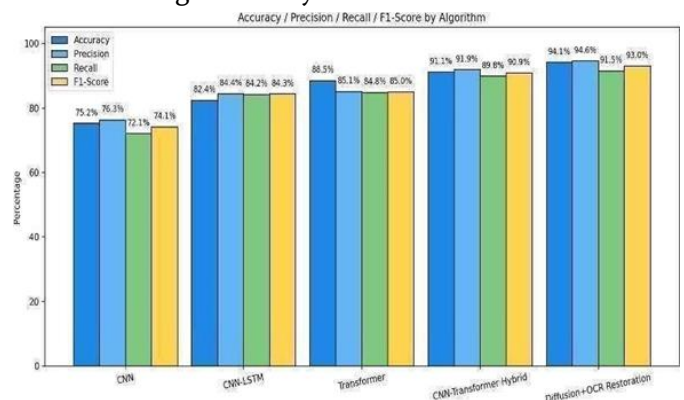


Fig. 5. Metric Evaluation of Algorithm on ModeTrans Dataset

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